

2.5

24. $UU^T = I$

$$U = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore U^{-1} = \begin{bmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{bmatrix}$$

Note: Problem differs. This is #40 for Edition S for textbook.

40. $E_1 = \begin{bmatrix} 1 & 0 & 0 & c \\ a & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$E_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$E_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$L = E_1 E_2 E_3 = (E_1 E_2) E_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & d & 1 & 0 \\ c & e & 0 & 1 \end{bmatrix}$

$E_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ b & d & 1 & 0 \\ c & e & 0 & 1 \end{bmatrix}$

OK all $a_{ij} (i,j) \in (1,4)$ that $a_{ij} \neq 0$ were copied into L.

2.6

21. $A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}$

$\Rightarrow \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix} \Rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & b-a & c-a & c-a \\ 0 & b-a & c-a & d-a \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & c-b & d-b \end{bmatrix}$

$U = \begin{bmatrix} a & a & a & a \\ 0 & b-a & b-a & b-a \\ 0 & 0 & c-b & c-b \\ 0 & 0 & 0 & d-b \end{bmatrix}$

$a \in \mathbb{R}$
 $a \neq b \neq c \neq d$

Didn't use eigenvectors

28. $A = LDU; A = L D U$

$\Rightarrow J^{-1} A J = I = U U^{-1}$
 $\Rightarrow L D U = L D U \Rightarrow L^{-1} L D U = L^{-1} L D U$
 $\Rightarrow U^{-1} L^{-1} L D U = D U U^{-1} \Rightarrow L^{-1} L D = D U U^{-1}$

$L = L^{-1} L D = I D U U^{-1}$
 $\Rightarrow D^{-1} L D = D L^{-1} D^{-1} \Rightarrow L = L$

$D = L^{-1} L D = D U U^{-1}$
 $\Rightarrow D = D \Rightarrow D = D$

$U = U^{-1} U I = D I U U^{-1} \Rightarrow U D = D I U$

$U D = D U$
 $U = U$

25. Unfortunately I do not have Matlab, but I coded a quick

Python

$J^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 \\ 1/3 & 2/3 & 1 & 0 & 0 \\ 1/6 & 1/3 & 1/3 & 1 & 0 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1 \end{bmatrix}$

26. Once again, I coded this in Python

$7K^{-1} = \begin{bmatrix} 6 & 5 & 4 & 3 & 2 & 1 \\ 5 & 4 & 3 & 2 & 1 & 0 \\ 4 & 3 & 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

multiplied by K

yields: $\begin{bmatrix} 7 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7 & 0 & 0 \\ 0 & 0 & 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 7 \end{bmatrix}$

$K = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix}$

Pattern: for each row before the pivot, the coefficients' numerator are incremented based on j, while i determines the denominator.

2.7

13. a. $P^3 = I$ ($P \neq I$) b. $\neq$ do not know what $\uparrow$ is

* Assuming 3 elements know what $\uparrow$ is

of permutations * Block Diagonal Matrix

$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $\hat{P} = \begin{bmatrix} 1 & 0 \\ 0 & P \end{bmatrix}$ $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow P^4 = I$

$\therefore 3$ permutations $\rightarrow$ simply go in cyclic fashion and will not return to the same spot.

#35 on version 5 (saved!)

$B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & 0 \end{bmatrix}$ B^T will also be northwest as the portion of zeros below the anti-diagonal are symmetric.

B^2 is not necessarily NW since there is always a row col such that $a_{ii} \cdot a_{ii} \neq 0$.

no #40 in v.s

39. $Q^T = Q^{-1}$ $Q = [q_1 \dots q_n]$ b. Given

If $Q^T = Q^{-1}$, $Q^T Q = I \Rightarrow S = Q$ for every q_i

$(\sqrt{(q_{1i})^2}) = 1 \Rightarrow \sqrt{(q_{1i})^2 + (q_{2i})^2 + \dots}$

a. This cannot be true as $\|q\|^2$ will always produce a value > 1 if any entry is nonzero. The only one that works is I .

$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & 1 \end{bmatrix}$, since there is only 1 nonzero entry (1) in each col, $\|q\|^2 = 1$

18. a. $M : A^T = A$

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 2 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}$ True

b. $M : A^T = -A$ True $\begin{bmatrix} 1 & 3 \\ 0 & 7 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 8 \end{bmatrix} \leftarrow A^T = A$

$\begin{bmatrix} 6 & 1 \\ -1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 12 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 6 & 13 \\ 1 & 6 \end{bmatrix}$ addition already

30 a. $S(S+T) = SS+ST$ since $S+T+T = S+2T$ Since multiplication is distributive, all combinations hold

b. $S+T$ produces another "line" while SUT is all lines formed with S and T . $SUT = S+T$ when $S=T$.

$P^2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$P^3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$P^4 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$

B^T , if it exists, will be southeast to fill in B to complete the identity matrix.

The shape of BC is not necessarily NW or SE, as it may or may not be filled with nonzero entries. This it is "filled".

b. Given $\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & 1 \end{bmatrix}$ it will

occur that the misalignment in diagonals (e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$) will always produce a det product such that $q_{1i} \cdot q_{1i} = 0$

c. $Q^T Q = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

$\cos^2 \theta + \sin^2 \theta = 1$

$\sin^2 \theta + \cos^2 \theta = 1 \Rightarrow q_{12} = q_{21} = \sin^2 \theta$

23. b. is Ex. 1: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 100 \\ 1 & 1 & -1 \end{bmatrix}$

Ex 2: $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & 6 \end{bmatrix}$

$Ax=b$ only solvable when $\mathbb{R}$ doesn't change as all possible b 's are already "formable" with A .

32. Matrices A and $[AB]$ all contain the same amt. of nonzero rows $\Rightarrow$ same space. $A = \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 6 \\ 0 & 0 \end{bmatrix}$... then A is an invertible matrix.